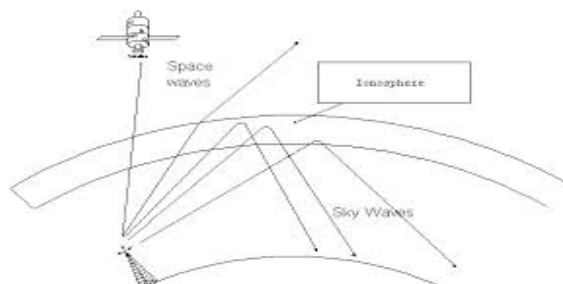


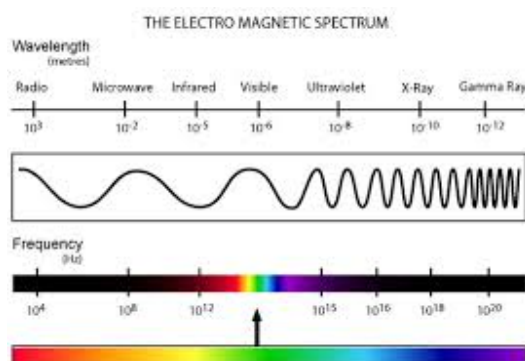


Q1. What is sky wave propagation?



Radio waves from the transmitting antenna reach the receiving antenna after reflection in the ionosphere, this wave propagation is called *sky wave propagation*.

Q2. Write the following radiations in ascending order in respect of their frequencies : X-rays, microwaves, UV rays and radio waves.



The ascending order of the frequencies of the radiations are : Radio waves, microwaves, UV rays and X-rays.

Q3. Magnetic field lines can be entirely confined within the core of a toroid, but not within a straight solenoid. Why?

If the currents are running strictly around the surface of the solenoid or torus, the field will be inside and strictly vanish outside. Since the magnetic field induction outside the toroid is zero.



Q4. You are given following three lenses. Which two lenses will you use as an eyepiece and as an objective to construct an astronomical telescope?

Lenses	Power (P)	Aperture (A)
1.1	3D	8 cm.
1.2	6D	1 cm.
1.3	10D	1 cm.

We know that $P=1/f$, $f=1/P$, for $P = 3$ $f=0.33$, For $P=10$, $f=0.1$, Magnifying power = f_o/f_e i.e for higher magnification $f_o > f_e$, if aperture is more it will allow more light to come in.

For a telescope, lens L_1 is used as objective as its aperture is largest (focal length is maximum). The lens L_3 is used as eye piece as its focal length is smaller.

Q5. If the angle between the pass axis of polarizer and the analyzer is 45° , write the ratio of the intensities of original light and the transmitted light after passing through the analyzer.

Given $\theta = 45^\circ \therefore \cos \theta = \cos 45^\circ = \frac{1}{\sqrt{2}}$

$$\therefore I = I_0 \cos^2 \theta$$

$$\therefore I = I_0 \left(\frac{1}{\sqrt{2}} \right)^2$$

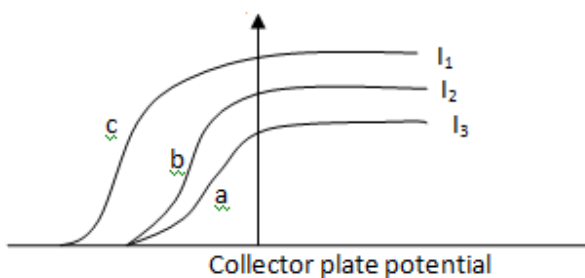
$$\frac{I}{I_0} = \frac{1}{2} \Rightarrow I : I_0 = 1 : 2$$



Q6. The figure shows a plot of three curves a, b, c, showing the variation of photocurrent vs. collector plate potential for three different intensities I_1 , I_2 and I_3 having frequencies ν_1 , ν_2 and ν_3 respectively incident on a photosensitive surface.

Point out the two curves for which the incident radiations have same frequency but different intensities.

Photoelectric current



The stopping potential V_s is same for a and b, hence frequencies of incident light are same for a and b curve, but intensities are different ($I_2 > I_1$).

Q7. What type of wavefront will emerge from a (i) point source, and (ii) distant light source?

(i) For the point source light touches the surface of sphere at the same time, since the point is near, curvature is more and type of wave front is Spherical wavefront. Point source $\rightarrow$ Spherical wave front.

(ii) If the source of light is at distant place, curvature is less as radius of curvature is more and it becomes a Plane wavefront. Distant Source $\rightarrow$ Plane wave front.

Q8. Two nuclei have mass numbers in the ratio 1 : 2. What is the ratio of their nuclear densities?

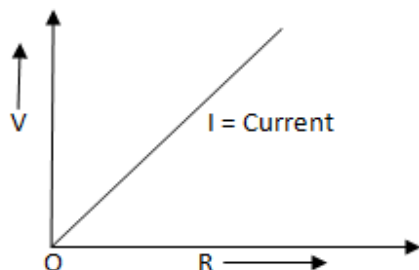
Nuclear density is the density of the nucleus of an atom, nuclear density is independent of mass number. Since the nuclear density is same for all nuclei. The ratio of their densities are 1 : 1.



Q9. A cell of emf 'E' and internal resistance 'r' is connected across a variable resistor 'R'. Plot a graph showing the variation of terminal potential 'V' with resistance R. Predict from the graph the condition under which 'V' becomes equal of 'E'.

Potential drop across external resistance increases if resistance increases however because of internal resistance there is also potential drop inside the battery, considered as wastage. If no external current flows through the circuit then potential drop inside batter (I r) becomes zero and Applied emf becomes equal to Potential drop across external resistance. Applied emf The graph between V and R is shown as

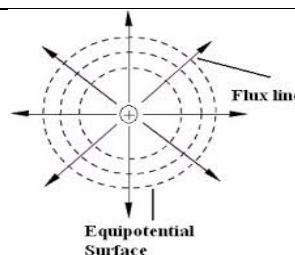
$V = E - Ir$, when $I=0$ then $V=E$



Q10.(I) Can two equi-potential surfaces intersect each other? Give reasons.

(II) Two charges $-q$ and $+q$ are located at points A (0, 0, -a) and B (0, 0, +a) respectively. How much work is done in moving a test charge from point P (7, 0, 0) to Q (-3, 0, 0)?

- (I) We know that $E = -dv/dr$, electric intensity is defined as potential gradient. In case two equipotential surfaces intersect at the point of intersection we can draw two tangents and this will give two directions of intensity at the single point, this is not possible, hence equipotential surface do not intersect.





(II) We know that Work done = Charge X Potential Difference = $q (V_2 - V_1)$

Potential at P (7, 0, 0) is

$$V_1 = \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(7-0)^2+0+(-a-0)^2}} + \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(7-0)^2+0+(a-0)^2}}$$

$$= \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{49+a^2}} + \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{49+a^2}} = 0$$

Potential at Q (-3, 0, 0) is

$$V_2 = \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(-3-0)^2+(-a)^2}} + \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(-3-0)^2+(a)^2}}$$

$$= \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{9+a^2}} + \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{9+a^2}} = 0$$

$$\therefore \text{Workdone} = q(V_2 - V_1) = q(0 - 0) = 0$$

Hence $w = 0$.

Q11. By what percentage will the transmission range of a TV tower be affected when the height of the tower is increased by 21%?

Ans. The distance (d) of TV coverage is given by

$$d = \sqrt{2hR} \dots \dots (i)$$

When height of the tower (h) is increased by 21%, $h_1 = h + 21h/100 = 1.21 h$

$$\therefore d_1 = \sqrt{2h_1R}$$

$$d_1 = \sqrt{2 \times 1.21 hR} = 1.1 \sqrt{2hR}$$

$$d_1 = 1.1 d = 1.10 d$$

$\therefore$ TV coverage is increased by 10%.

Q12. Derive an expression for drift velocity of free electrons in a conductor in terms of relaxation time.

Ans. **Expression for drift velocity:**

Let V = potential difference applied across the ends of the conductor

L = length of the conductor

m = mass of an electron then magnitude of electric field E is $E = \frac{V}{L}$

The direction of this electric field is from positive end to negative end of the conductor. Since the charge on the electron is (-e), then the force experienced in the conductor is

$$\vec{F} = -e \vec{E} \dots \dots (i)$$



Hence acceleration of each electron

$$m \vec{a} = -e \vec{E} \Rightarrow \vec{a} = \frac{e \vec{E}}{m} \dots \dots \dots (ii)$$

$$[\because F = ma]$$

Due to this acceleration, the free electron apart from this thermal velocity, acquires additional velocity component in a direction opposite to the direction of electric field.

At any instant of time, the velocity acquired by electron having thermal velocity $\vec{u}_1$ is

$$\vec{v}_1 = \vec{u}_1 + \vec{a} \tau_1$$

Where, τ_1 is the time elapsed. Similarly, the velocities acquired by other electrons in the conductor is

$$\vec{v}_2 = \vec{u}_2 + \vec{a} \tau_2 \text{ and so on.}$$

$$\therefore \vec{v}_n = \vec{u}_n + \vec{a} \tau_n$$

The average velocity of all the free electrons in the conductor is the drift velocity $\vec{v}_d$ of free electrons.

$$\therefore \vec{v}_d = \frac{\vec{v}_1 + \vec{v}_2 + \dots + \vec{v}_n}{n}$$

$$= \frac{(\vec{u}_1 + \vec{a} \tau_1) + (\vec{u}_2 + \vec{a} \tau_2) + \dots + (\vec{u}_n + \vec{a} \tau_n)}{n}$$

$$= \left(\frac{\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_n}{n} \right) + \vec{a} \left(\frac{\tau_1 + \tau_2 + \dots + \tau_n}{n} \right)$$

$$= 0 + \vec{a} \tau = \vec{a} \tau \dots \dots \dots (iii)$$

[Since average thermal velocity of electrons is zero.]

Where, $\tau = \frac{\tau_1 + \tau_2 + \dots + \tau_n}{n}$ is called relaxation time. Its value is of the order of 10^{-14} seconds.

Putting the value of $\vec{a}$ in (iii), we get

$$\vec{v}_d = \frac{-e \vec{E}}{m} \tau$$

Hence average drift speed, $v_d = \frac{eE}{m} \tau$

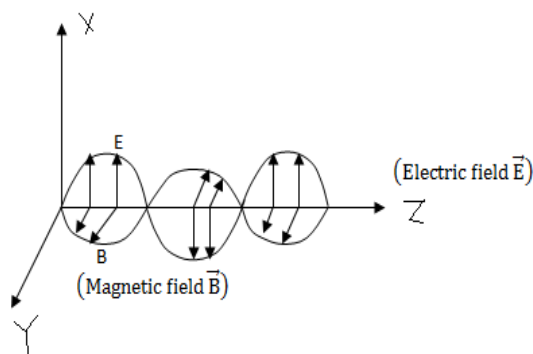


Q13. How does a charge q oscillating at certain frequency produce electromagnetic waves?

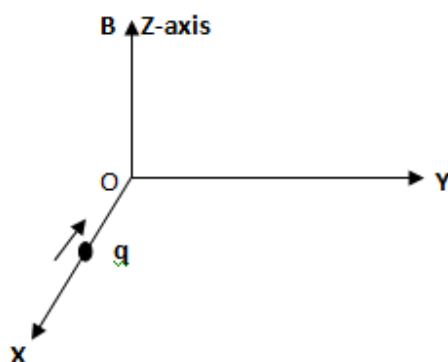
Sketch a schematic diagram depicting electric and magnetic fields for an electromagnetic wave propagating along the Z-direction.

Surrounding an electric charge we have electric field, if the charge oscillates then electric field changes which creates magnetic field also. Hence accelerated charge produces electric and magnetic fields, the electric and magnetic field vectors are mutually perpendicular and direction of propagation of the electromagnetic wave is perpendicular to the plane containing electric and magnetic vector.

Schematic Diagram



Q14. A charge ' q ' moving along the X-axis with a velocity $\vec{v}$ is subjected to a uniform magnetic field B acting along the Z-axis as it crosses the origin O .

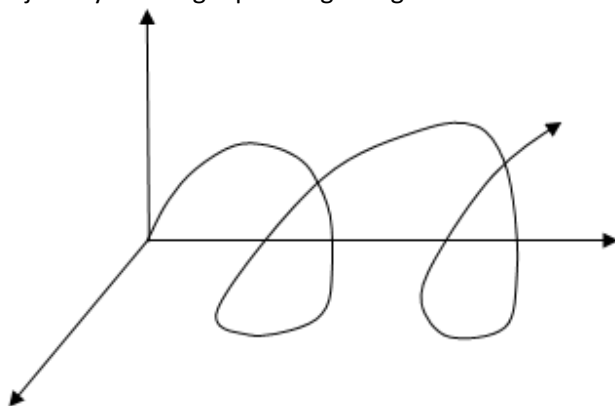


Trace its trajectory.

- (i) Does the charge regain kinetic energy as it enters the magnetic field? Justify your answer.

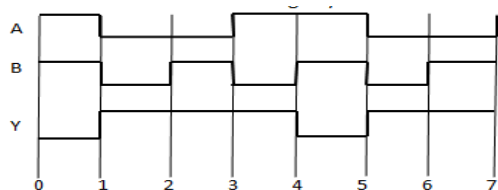


- (i) The trajectory of charge q moving along x -axis will be **helical**.



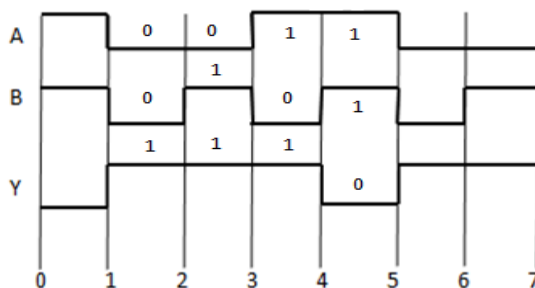
- (ii) The speed and kinetic energy of the particle remain constant but the velocity of the charged particle changes its direction.

Q15. The following figure shows the input waveforms (A, B) and the output waveform (Y) of gate. Identify the gate, write its truth table and draw its logic symbol.



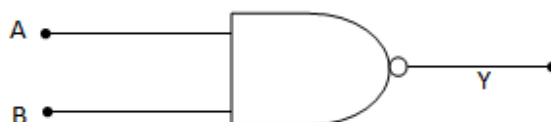
- (i) Gate : NAND Gate Truth Table

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



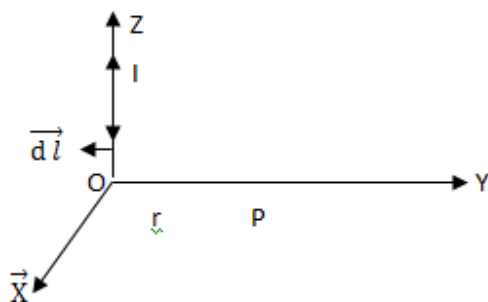


(ii) Logic symbol :



Q16. State Biot-Savart law.

A current flows I flows in a conductor placed perpendicular to the plane of the paper. Indicate the direction of the magnetic field due to a small element $d\vec{l}$ at point P situated at a distance $\vec{r}$ from the element as shown in the figure.





Given

i = current flowing through the element

dl = the length of the element of the conductor

$r = AP$ = the distance of the given point from the element of the conductor

$\hat{r}$ = a unit vector along AP

$\hat{dl}$ = a vector of unit length along the axis of the element in the direction of current

θ = the angle between $\hat{r}$ and $\hat{dl}$

Let $\vec{db}$ = magnetic induction vector at the point P due to the current element

Experimentally it has been found that

$$|\vec{db}| \propto i, |\vec{db}| \propto dl, |\vec{db}| \propto \frac{1}{r^2}, |\vec{db}| \propto \sin \theta$$

$$db \propto \frac{idl \sin \theta}{r^2}$$

$$db = K' \frac{idl \sin \theta}{r^2} \rightarrow (1)$$

Where K' is a constant of proportionality. The value of K' depends on

(1) Choice of unit

(2) Nature of surrounding medium

In SI system db in tesla, i in ampere, r in meter for free space (air or vacuum)

$$K' = 10^{-7} = \frac{\mu_0}{4\pi} \text{ where } \mu_0 = 4\pi \times 10^{-7} \text{ henry/meter} = \text{permeability in free space}$$

$$db = \frac{\mu_0}{4\pi} \frac{idl \sin \theta}{r^2} \rightarrow (2)$$

Equation (2) which gives magnitude of magnetic field at P is known as Laplace Formula.

$$db = \frac{\mu_0}{4\pi} \frac{i}{r^2} \left(\vec{dl} \times \hat{r} \right) \rightarrow (3)$$

Biot-Savart law : According to this law, the magnitude of magnetic field induction at a point due to a current element of length dl , carrying current i , at a distance r from the element is given by

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{i dl \sin \theta}{r^2}$$

$$\text{Vector form : } |d\vec{B}| = \frac{\mu_0}{4\pi} \cdot \frac{|dl \times \hat{r}|}{r^2}$$

The direction of magnetic field $d\vec{B}$ is perpendicular to the plane containing $d\vec{l}$ and $\hat{r}$ and is directed **inward**.



Q17. Why are high frequency carrier waves used for transmission?

Ans. Reason

1. High frequency carrier waves travel through space or medium with the speed of light and
2. High frequency carrier waves are not obstructed by earth's atmosphere.

Or, What is meant by term 'modulation'? Draw a block diagram of a simple modulator for obtaining an AM signal.

Modulation is a process of mixing a signal with a sinusoid to produce a new signal. It is the phenomenon of superimposing the low frequency message signal on a high frequency wave (carrier wave).

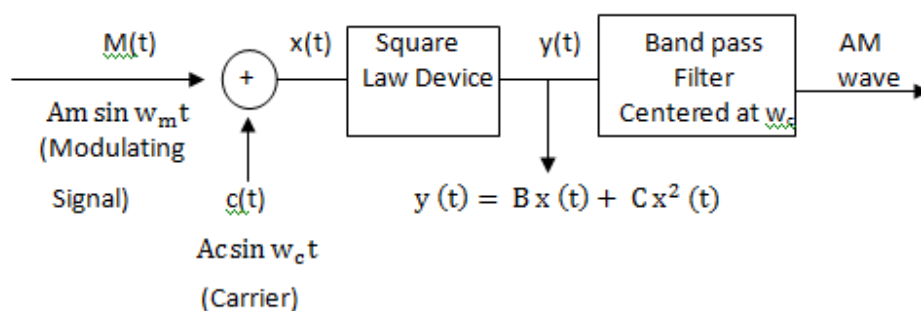
There are 3 basic types of modulation: Amplitude modulation, Frequency modulation, and Phase modulation.

In Amplitude modulation the amplitude of the carrier signal is modulated (changed) in proportion to the message signal while the frequency and phase are kept constant.

In frequency modulation the frequency of the carrier signal is modulated (changed) in proportion to the message signal while the amplitude and phase are kept constant.

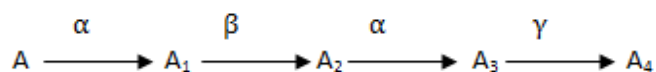
In phase modulation the phase of the carrier signal is varied accordance to the low frequency of the message signal is known as phase modulation.

Block diagram of simple modulator:





Q18. A radio active nucleus 'A' undergoes a series of decays according to the following scheme:

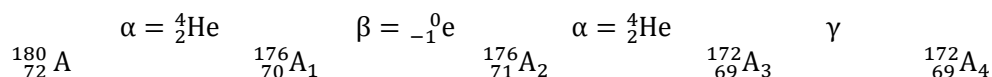


The mass number and atomic number of A are 180 and 72 respectively. What are these numbers for A_4 ?

Answer : We know that alpha particle when emits atomic mass decreased by 4 unit and atomic number decreased by 2 unit, while Beta emission increases atomic number but does not bring any changes to mass.

Gama emission does not bring any changes neither in atomic mass nor in atomic number.

Alpha = doubly ionized helium, Beta = electron, Gama = Electro Magnetic wave.



$\therefore$ The mass number of $A_4 = 172$
and the atomic number of $A_4 = 69$

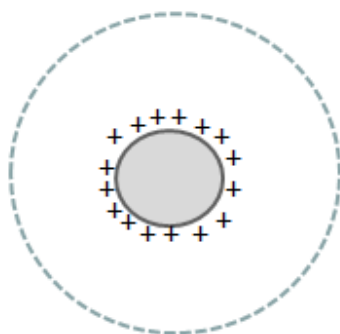
Q19. A thin conducting spherical shell of radius R has charge Q spread uniformly over its surface. Using Gauss's law, derive an expression for an electric field at a point outside the shell.

Draw a graph of electric field $E(r)$ with distance r from the centre of the shell for $0 \leq r \leq \infty$.



Applications of Gauss's Theorem :

To find intensity at a point due to a uniformly charged spherical shell



Q = charge on the surface of the shell

R = Radius of the shell

$r = OP$ = distance of the given point from the centre of the shell.

ϵ = Permittivity of the medium

Let E = Electric intensity at P due to the charged shell.

$$E = \frac{1}{4\pi\epsilon} \frac{Q}{r^2} \rightarrow (1)$$

Equation (1) is applicable when the given charge is point charge but in this case since the charged shell can not be treated as point charge we can not apply this formula.

Let us imagine a concentric shell of radius r passing through P this is our Gaussian surface.

Surface area of the Gaussian surface $A = 4\pi r^2$

Since the lines of force leave the surface of a charged body perpendicularly hence the lines of force are all radial to the charged sphere and hence force are all radial to the charged sphere and hence they are also radial to the Gaussian surface and cutting the surface perpendicularly.

Using definition of flux through the Gaussian surface $\phi = EA = E \cdot 4\pi r^2 \rightarrow (1)$



Case I : Let the point P lie outside the charged shell $OP = r > R$

Applying the Gauss's theorem the flux through the Gaussian surface

$$\phi = \frac{\text{Charged enclosed}}{\epsilon} = \frac{Q}{\epsilon} \rightarrow (2)$$

Equating (1) and (2):

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon}$$

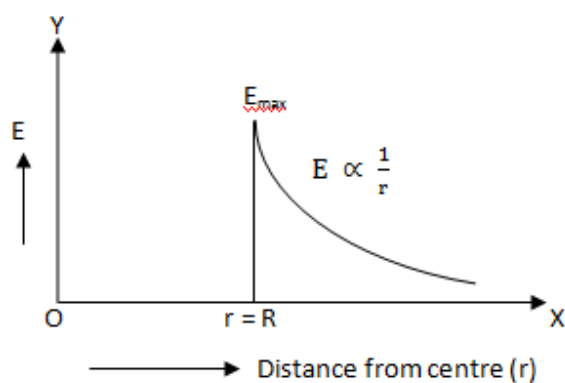
$$E = \frac{1}{4\pi\epsilon} \frac{Q}{r^2} \rightarrow (3)$$

If we imagine a point charge Q at the center of the shell intensity at P

$$E = \frac{1}{4\pi\epsilon} \frac{Q}{r^2} \rightarrow (4)$$

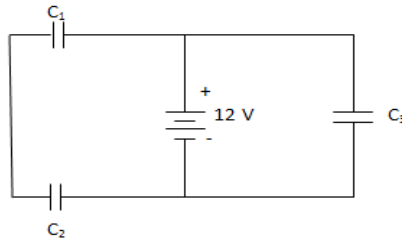
Thus comparing equation (3) & (4) we can say that to find potential and intensity at any external point due to a charged shell the shell can be assumed to be concentrated at its centre.

Graphical Presentation: The variation of the electric field intensity $E(r)$ with distance r from the centre for shell $0 \leq r \leq \infty$ is shown below:





Q20. Three identical capacitors C_1 , C_2 and C_3 of capacitance $6\ \mu\text{F}$ each are connected to a $12\ \text{V}$ battery as shown.



Find

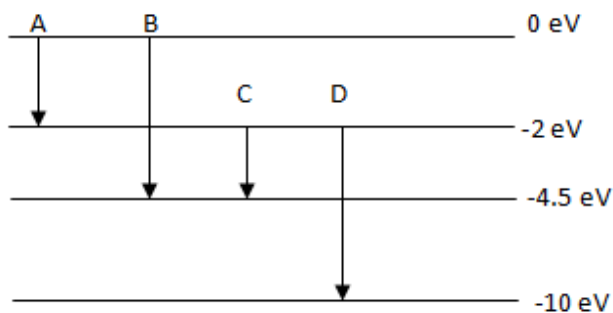
- (i) Charge on each capacitor
- (ii) Equivalent capacitance of the network
- (iii) Energy stored in the network of capacitors

Ans.

- (i) Here $V = 12\ \text{V}$ and $C_1 = C_2 = C_3 = 6\ \mu\text{F} = 6 \times 10^{-6}\ \text{F}$
 Charge on capacitor C_3 is
 $q_3 = C_3 V = 6 \times 10^{-6} \times 12 = 72 \times 10^{-6}$
 $= 72\ \mu\text{C}$
 Since capacitor C_1 and C_2 are in series
 $\therefore$ Equivalent capacitance $\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2}$
 $\frac{1}{C_S} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$
 $\therefore C_S = 3\ \mu\text{F}$
 Charge on capacitor C_1 and C_2 is
 $q = C_S V = 3 \times 10^{-6} \times 12 = 36 \times 10^{-6} = 36\ \mu\text{C}$
 $\therefore$ Charge on each capacitor C_1 and C_2 is $36\ \mu\text{C}$
- (ii) Since capacitor C_1 and C_2 are in series
 $\therefore$ Equivalent capacitance $C_S = 3\ \mu\text{F}$
 Now C_3 and C_S are in parallel
 $\therefore$ Equivalent capacitance $C = C_3 + C_S$
 $= 6 + 3 = 9\ \mu\text{F}$
- (iii) Energy stored
 $= \frac{1}{2} CV^2$
 $= \frac{1}{2} \times 9 \times 10^{-6} \times (12)^2$
 $= \frac{1}{2} \times 9 \times 10^{-6} \times 144 = 648 \times 10^{-6}$
 $= 6.48 \times 10^{-4}\ \text{J}$



Q21. (a) The energy levels of an atom are as shown below. Which of them will result in the transition of a photon of wavelength 276 nm?



(b) Which transition corresponds to emission of radiation of maximum wavelength?

Ans. (a) **Considering element A**

Ground state energy = $E_1 = -2 \text{ eV}$ and Energy of the Excited state = $E_2 = 0 \text{ eV}$

Difference of Energy States $E = E_2 - E_1 = 0 - (-2) = 2 \text{ eV}$

$$\therefore \text{Wavelength of photon emitted, } \lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2 \times 1.6 \times 10^{-19}} = \frac{19.878 \times 10^{-7}}{3.2} \\ = 6.211 \times 10^{-7} \text{ m} = 621.1 \text{ nm.}$$

Considering element B

$E_1 = -4.5 \text{ eV}$, $E_2 = 0 \text{ eV}$ Difference in energy states = $E = E_2 - E_1 = 0 - (-4.5) = 4.5 \text{ eV}$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{4.5 \times 1.6 \times 10^{-19}} = \frac{19.878 \times 10^{-7}}{7.2} = 2.760 \times 10^{-7} \text{ m} = 276 \text{ nm.}$$

Considering element C

$E_1 = -4.5 \text{ eV}$, $E_2 = -2 \text{ eV}$ Difference in energy states = $E = E_2 - E_1 = -2 - (-4.5) = 2.5 \text{ eV}$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2.5 \times 1.6 \times 10^{-19}} = \frac{19.878 \times 10^{-7}}{4} = 4.969 \times 10^{-7} \text{ m} = 496.9 \text{ nm.}$$

Considering element D

$E_1 = -10 \text{ eV}$, $E_2 = -2 \text{ eV}$ Difference in energy stages = $E = E_2 - E_1 = -2 - (-10) = 8 \text{ eV}$

$$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{8 \times 1.6 \times 10^{-19}} = \frac{19.878 \times 10^{-7}}{12.8} = 1.552 \times 10^{-7} \text{ m} = 155.2 \text{ nm.}$$

$\therefore$ Element B has a photon of wavelength 276 nm.

(b) From the above results we find element A emits maximum wavelength = 621 nm.



Q22. A proton and an alpha particle are accelerated through the same potential. Which one of the two has (i) greater value of de-Broglie wavelength associated with it, and (ii) less kinetic energy? Justify your answers.

Answer

We know that de-Broglie wavelength λ associated with potential V is $\lambda = \frac{h}{\sqrt{2meV}}$

$$\therefore \lambda \propto \frac{1}{\sqrt{m}} \quad [\text{For } V \text{ (potential) is same}]$$

Since mass of alpha particle is larger than a proton $\therefore \lambda_{\text{proton}} > \lambda_{\text{alpha}}$, Proton will be associated with greater de-Broglie wavelength.

$$(ii) \text{ Since } KE = hv = h \frac{c}{\lambda}$$

$$E \propto \frac{1}{\lambda} \text{ Since } \lambda_{\text{proton}} > \lambda_{\text{alpha}}$$

$\therefore$ Kinetic energy of proton (K_{proton}) is less than Kinetic Energy of Alpha (K_{alpha})

Q23. In a single slit diffraction experiment, when a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. Explain why?

State two points of diffraction between the interference patterns obtained in Young's double slit experiment and the diffraction pattern due to a single slit.

Answer

The bright spot is due to constructive interference of waves due to diffraction from the edge of the circular obstacle.

Difference between Interference and Diffraction fringes pattern :

Pattern in Interference	Pattern in Diffraction
1. All bright fringes are equally bright, width of the fringes is equal in interference	1. The brightness of successive bright fringes from the centre goes on decreasing, width of fringes are not equal.
2. The region of minimum intensity is perfectly dark in interference.	2. In diffraction region of minimum intensity are not perfectly dark.



Q24. (a) Define self inductance. Write its SI units.

(b) Derive an expression for self inductance of a solenoid of length l cross sectional area A having n number of turns.

Answer

Self inductance (or coefficient of self induction) : Self inductance of a coil is numerically equal to the amount of magnetic flux linked with the coil when unit current flows through the coil. Its SI unit is henry (H).

Self inductance of a long solenoid: Considering a long solenoid.

l = length of solenoid,

N = number of turns

r = radius

I = current flowing through solenoid

B = Magnetic field set up in the coil is

$$\phi = \vec{B} \cdot \vec{A} = BA \cos 0 = BA$$

$$\phi = \mu_0 \frac{Ni}{l} \pi a^2 \rightarrow (1)$$

Since the flux is varying the induced emf across the coil at an instant t

$$e = -N \frac{d\phi}{dt} \text{ (from Faradays laws of electromagnetic induction)}$$

$$\text{Using equation(1): } e = -N \frac{\mu_0 N \pi a^2}{l} \frac{di}{dt}$$

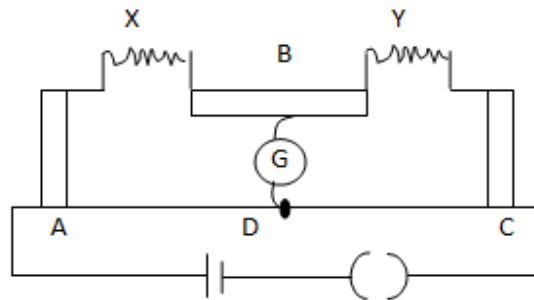
$$\text{Putting } \frac{\mu_0 \pi N^2 a^2}{l} = L = \text{Constant of the coil} = \text{Coefficient of self induction}$$

$$e = -L \frac{di}{dt} \quad \left\{ L = \frac{\text{Henry}}{m} \frac{m^2}{m} = \text{Henry} \right.$$

$$\therefore \text{ Self inductance, } L = \frac{\phi}{i} = \frac{\mu_0 N^2 A}{l}$$



Q25.



The figure shows experimental set up of a meter bridge. When the two unknown resistances X and Y are inserted, the null point D is obtained 40 cm from the end A. when the resistance of $10\ \Omega$ is connected in series with X, the null points shifts by 10 cm. Find the position of the null point when the $10\ \Omega$ resistance is instead connected in series with resistance 'Y'. Determine the values of the resistances X and Y.

Answer: In Meter bridge ABCD forms Wheatstone bridge, so the resistance in first(X) and 2nd (Y) arm should be equal to the ratio of 3rd (AD) and 4th Arm(DC).

At the null point at D

$$\frac{X}{Y} = \frac{AD}{DC}$$

$$\frac{X}{Y} = \frac{40}{100-40} \Rightarrow \frac{X}{Y} = \frac{40}{60}$$

$$\frac{X}{Y} = \frac{2}{3} \Rightarrow X = \frac{2}{3}Y \dots \dots \dots (1)$$

When a resistance of $10\ \Omega$ is connected in series with X.

$\therefore$ At the new null point

$$\frac{X+30}{Y} = \frac{50}{100-50} \Rightarrow \frac{X+30}{Y} = \frac{50}{50} = 1 \therefore X + 30 = Y \dots \dots \dots (2)$$

Putting the value of X, we get

$$\frac{2}{3}Y + 30 = Y$$

$$30 = Y - \frac{2}{3}Y \Rightarrow 30 = \frac{1}{3}Y \Rightarrow Y = 90\ \Omega$$

Putting the value of Y in (1), we get

$$X = \frac{2}{3} \times 90 = 60\ \Omega$$

When the $10\ \Omega$ resistance is connected in series with Y then

$$\frac{X}{Y+30} = \frac{60}{90+30} = \frac{60}{120} = \frac{60}{100+20} \therefore \text{Hence null point is shifted 20 cm towards right.}$$



Q26. Derive the expression for force per unit length between two long straight parallel current carrying conductors. Hence define one ampere.

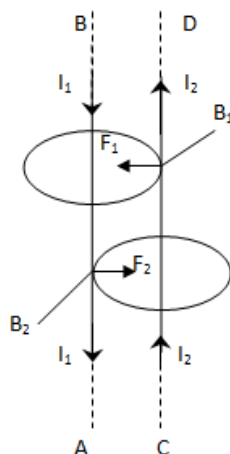
Answer:

Force between two parallel current carrying wires

Given: AB and CD are two infinitely long conductors, placed parallel to each other

r = distance of separation between two wires.

I_1 and I_2 = current flowing through AB and CD respectively (direction of current taken same).



Magnetic field produced by current I_1 at any point on CD is

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$

This field acts perpendicular to CD and into the plane of paper. It exerts a force on wire CD carrying current I_2 is

$$F = B_1 I_2 l$$

$$F = \frac{\mu_0}{2\pi r} \times I_1 \cdot I_2 l$$

$$F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r} \cdot l$$

$\therefore$ Force per unit length is

$$F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r}$$

$$\therefore F = \frac{\mu_0}{4\pi} \cdot \frac{2I_1 I_2}{r}$$

By Fleming's left hand rule, this force will be attractive.

Definition of ampere :

Let $I_1 = I_2 = 1$ A, $r = 1$ m then



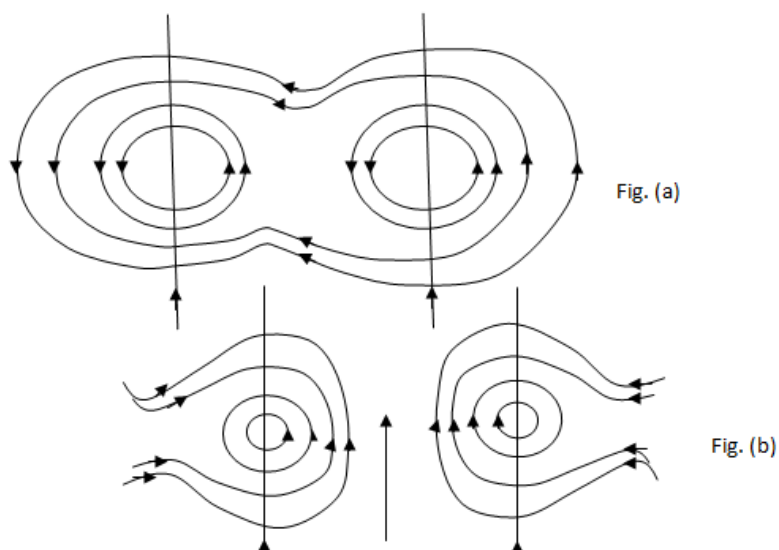
$$F = \frac{4\pi \times 10^{-7} \times 2 \times 1 \times 1}{4\pi \times 1}$$

$$[\because \mu_0 = 4\pi \times 10^{-7}]$$

$$F = 2 \times 10^{-7} \text{ Nm}^{-1}$$

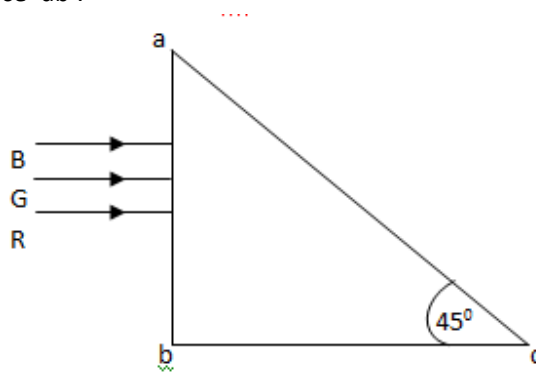
Thus, one ampere can be defined as the amount of current which when flows through each of the two parallel uniform long linear conductors placed in free space at a distance of one metre from each other will attract or repel each other with a force of 2×10^{-7} N per metre of their length.

Magnetic field Pattern: The pattern of the magnetic field indicates the force of attraction between two parallel conductors carrying current in the same direction (Fig. a) and force of repulsion between two parallel conductors carrying current in opposite direction (Fig. b).



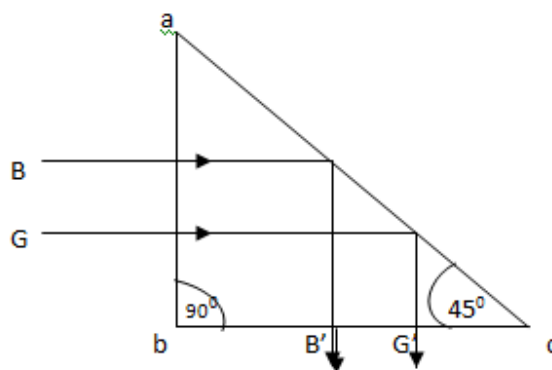


Q27. Three light rays red (R), green (G) and blue (B) are incident on a right angled prism 'abc' at face 'ab'. The refractive indices of the material of the prism for red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. Out of the three which colour ray will emerge out of face 'ac'? Justify your answer. Trace the path of these rays after passing through face 'ab'.



Answer:

The red light ray (R) will emerge out of face. The path of green (G) and blue (B) light rays will be as



$\mu = A + \frac{B}{\lambda^2}$, we know wave length of Red (R) is maximum; hence refractive index of this prism for red will be minimum. Also we have $\mu_1 \sin i = \mu_2 \sin r$, and critical angle for red is maximum hence except red colour Green and Blue light rays will incident at ac surface more than the critical angle hence blue and green will suffer total internal reflection whereas Red will emerge out for ac face.



- Q28.** (a) Derive an expression for the average power consumed in a series LCR circuit connected to a.c. source in which the phase difference between the voltage and the current in the circuit is ϕ .
 (b) Define the quality factor in an a.c. circuit. Why should the quality factor have high value in receiving circuits? Name the factors on which it depends.

Answer

(a) **To find power consumed in LCR circuit :**

$$\text{Given } E = \text{alternating emf applied to an LCR circuit} = E_0 \sin \omega t \dots \dots (1)$$

If alternating current developed lags behind the applied emf by a phase angle ϕ then

$$I = I_0 \sin(\omega t - \phi) \dots \dots (2)$$

Total work done over a complete cycle is

$$\begin{aligned} W &= \int_0^T EI \, dt \\ &= \int_0^T E_0 \sin \omega t \, I_0 \sin(\omega t - \phi) \, dt \\ &= E_0 I_0 \int_0^T \sin \omega t \sin(\omega t - \phi) \, dt \\ &= \frac{E_0 I_0}{2} \int_0^T 2 \sin \omega t \sin(\omega t - \phi) \, dt \\ &= \frac{E_0 I_0}{2} \int_0^T [\cos(\omega t - \omega t + \phi) - \cos(\omega t + \omega t - \phi)] \, dt \\ &\quad [\because 2 \sin A \sin B = \cos(A - B) - \cos(A + B)] \\ &= \frac{E_0 I_0}{2} \int_0^T [\cos \phi - \cos(2\omega t - \phi)] \, dt \\ &= \frac{E_0 I_0}{2} \left[t \cos \phi - \frac{\sin(2\omega t - \phi)}{2\omega} \right]_0^T \\ &= \frac{E_0 I_0}{2} [T \cos \phi] \end{aligned}$$

$\therefore$ Average power in LCR circuit over a complete cycle is

$$P = \frac{W}{T} = \frac{E_0 I_0}{2} \cos \phi = \frac{E_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cdot \cos \phi$$

$$\therefore P = E_V I_V \cdot \cos \phi$$

(b) **Quality factor in an a.c. :**

The Q-factor of a resonant LCR circuit is defined as ratio of the voltage drop across inductor or capacitor to the applied voltage.

$$\therefore Q = \frac{\text{Voltage across L or C}}{\text{applied voltage}}$$

Since $V_L = I X_L$ and $V = IR$

$$\therefore Q = \frac{I X_L}{IR} = \frac{\omega_0 L}{R}$$

Hence at high frequencies, the Q-factor $\frac{\omega_0 L}{R}$ is quite large.



∴ The voltage drop across the inductor will be quite large as compared to the applied voltage.

$$\text{Also } Q = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} \quad \left[\because \omega_0 = \frac{1}{\sqrt{LC}} \right]$$

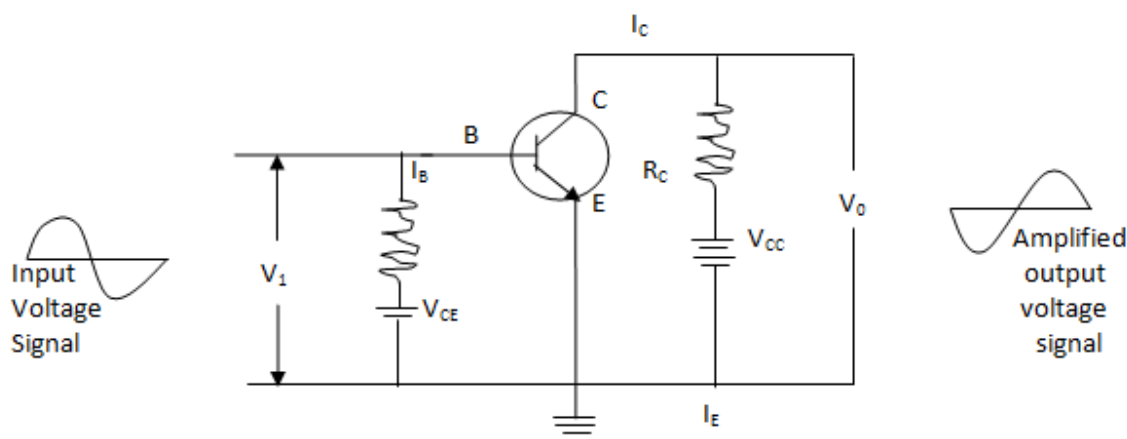
$$\therefore Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

The Q-factor of LCR-series circuit depends on L, C and R.

Q29. Explain, with the help of a circuit diagram, the working of n-p-n transistor as a common emitter amplifier.

Answer:

Transistor as an Amplifier - Common Emitter (C E) configuration: The circuit diagram of a common emitter amplifier using n-p-n transistor is given below.



The input (base-emitter) circuit is forward biased and the output (Collector-emitter) circuit is reversed biased.

When no a.c. signal is applied, the potential difference V_{CC} between the collector and emitter is given by

$$V_{CC} = V_{CE} + I_C R_C$$

When an a.c. signal is fed to the input circuit, the forward bias increases during the positive half cycle of the input. This results in increase in I_C and decrease in V_{CC} . Thus during positive half cycle of the input, the collector becomes less positive.

During the negative half cycle of the input, the forward bias is decreased resulting in decrease in I_E and hence I_C . Thus V_{CC} would increase making the collector more positive.

Hence in a common-emitter amplifier, the output voltage is 180° out of phase with the input voltage.



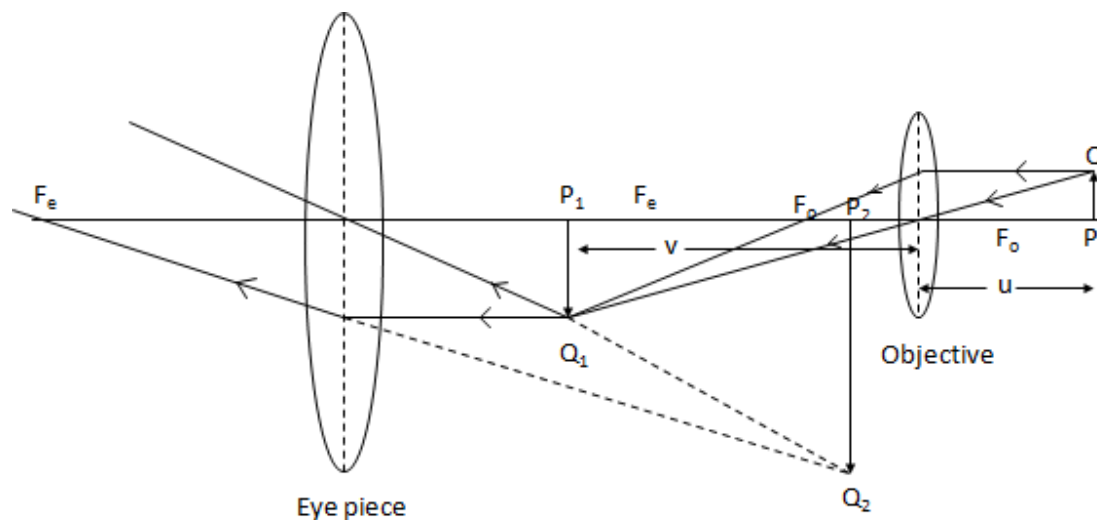
- (i) Input signal voltage $V_i = I_B R_B$
 (ii) Output signal voltage $V_o = I_C R_C$
 (iii) Voltage gain (A_v) of the amplifier is the ratio of output voltage to the input voltage.

$$A_v = \frac{V_o}{V_i} = \frac{I_C R_C}{I_B R_B} = \beta \left(\frac{R_C}{R_B} \right) \therefore \beta = \frac{I_C}{I_B}$$

Q30. Draw the labeled ray diagram for the formation of image by a compound microscope. Derive the expression for the total magnification of a compound microscope. Explain why both the objective and the eyepiece of a compound microscope must have short focal lengths.

Answer

It consists of objective and eye piece. The objective is a convex lens of small aperture and small focal length. The aperture of the eye piece lens is slightly greater and the focal length is also small.



Ray diagram : PQ is an object kept in front of objective ($f_o < u < 2 f_o$), P_1Q_1 is the image formed by objective this image serves as object for eye piece, since P_1Q_1 lies between the focus and optical center of eye piece hence an extended image P_2Q_2 is formed by eye piece. Thus P_2Q_2 is the final image of PQ produced by the compound microscope.

Magnifying power = Height of final image / Height of original image = $P_2Q_2 / PQ = (P_1Q_1 / PQ) \times (P_2Q_2 / P_1Q_1)$

From the above figure = P_1Q_1 / PQ = Magnifying power of objective = M_o

P_2Q_2 / P_1Q_1 = Magnifying power of eye piece = M_e



Magnifying power of compound microscope :

$$M = M_o \times M_e$$

For normal vision

$$M_o = \frac{v}{u} \frac{D}{f_e}$$

For distinct vision

$$M_o = \frac{v}{u} \left(1 + \frac{D}{f_e} \right)$$

For large magnifying power, f_o and f_e both have to be small. Also f_o is taken to be smaller than f_e so that the field of view may be increased.